TEXORA · LECTURE SHEET

Pattern Making

How body measurements become the paper (or digital) templates that govern cutting: the principles of drafting, the shirt and trouser pattern sets step by step, and grading — the controlled enlargement that produces the full size range.

TE-2207 · APPAREL MANUFACTURING-1 · TOPIC 3 · LEVEL 2 TERM 2

1. What a pattern is and does

A **pattern** is the flat template of one garment component, drawn to exact dimensions, from which the fabric pieces are cut. The complete set of patterns for a style is the factory's master instruction for shape: the cutter cuts what the pattern says, the machinists sew what was cut, and the finished garment can only be as correct as the pattern set. Pattern making is therefore the point where the spec sheet (Topic 2) becomes physical.

Patterns exist in a hierarchy. The **basic block (sloper)** is a plain, close-fitting pattern drafted directly from body measurements, with no style features — the starting shape for a family of garments. The **working pattern** is developed from the block: style lines, fullness, openings and trims are added. The **production pattern** is the final, checked version used in the cutting room, carrying all allowances and markings; it is this set that is graded into sizes and laid into markers. Patterns are traditionally cut from pattern paper or card; in CAD systems the same hierarchy exists as digital files (Topic 7).

2. The two routes to a pattern

METHOD	HOW IT WORKS	WHERE IT RULES
Flat pattern (drafting)	The pattern is drawn in two dimensions on paper from measurements, using geometry, standard proportions and drafting formulas; style changes are made by cutting and spreading or closing parts of the draft (slash-and-spread / pivot methods).	Industrial production of woven garments — shirts, trousers, jackets — where repeatability and grading matter.
Draping (modelling)	Fabric or muslin is pinned and shaped directly on a dress form (stand) until the three-dimensional design is achieved; the	High-fashion and ladies' wear with soft, sculptural designs that cannot be described by flat geometry.

	draped pieces are then flattened, trued and transferred to paper.

Industrial apparel — the subject of this course — is overwhelmingly flat-pattern work, and the rest of this sheet follows the drafting route.

3. Tools of the pattern maker

TOOL	USE
Pattern paper / card	The drafting medium; card for production patterns that must survive repeated use.
L-square and straight rulers	Right angles and the long straight lines of the draft's construction frame.
French curve and hip curve	Smooth curved lines — armholes, necklines, side seams at the hip.
Measuring tape and protractor	Body/spec measurements and the angles of darts and pleats.
Notcher	Punches the small rectangular notches that mark matching points on pattern edges.
Awl / drill	Marks internal points (pocket positions, dart ends) as small holes inside the piece.
Tracing wheel	Transfers lines from one layer or pattern to another.
Pattern weights and shears	Hold paper flat; cut the finished pattern cleanly.

4. Principles of pattern making

Five principles govern every draft, whatever the garment.

4.1 Draft from measurements, in a construction order

The draft begins with a rectangle or frame built from the largest measurements (length × half chest for a shirt body), inside which construction lines — chest line, waist line, hip line — are fixed at proportions of the measurements. Details are added in a fixed order: outline, style lines, openings, then the separate pieces are traced off. Skipping the order produces pieces that no longer agree with each other.

4.2 Add allowances after the net shape

The drafted line is the **sewing (net) line**. Around it the pattern carries:

- **Seam allowance** — fabric beyond the sewing line so seams can be joined: typically 1 cm on most seams, 1.5 cm where strength or adjustment is needed (side seams, centre back), 0.6–0.75 cm on enclosed curved seams (collar, cuff) to reduce bulk.
- **Hem allowance** — the turn-up at the bottom: about 2–2.5 cm for a shirt hem, 3–4 cm for trouser bottoms.
- **Ease** — already inside the net shape (Topic 2): the draft is drawn to garment measurements, not body measurements.

4.3 Respect the grain

Every piece carries a **grain line** — an arrowed line that must lie parallel to the fabric's warp (lengthwise grain) unless the design states otherwise. A piece cut off-grain will twist on the body and hang badly; the grain line is the pattern's instruction to the marker maker (Topic 4) about orientation.

4.4 Mark everything the next operations need

- **Notches** — small cuts on edges marking where pieces join (single notch = front, double = back, by common convention) and where fullness is distributed.
- **Drill holes** — internal marks for pockets, darts and button positions.
- **The information block** — written on every piece: style number, piece name, size, number to cut (×1, ×2, pair), and grain line. An unmarked pattern is an accident waiting in the cutting room.

4.5 Walk the seams before release

Seams that will be sewn together are **walked** — measured against each other along their sewing lines. Front and back side seams, sleeve cap against armhole (allowing the small designed cap ease), collar against neckline: the lengths must agree within their designed ease. Pattern checking also confirms the set is complete and mirrored pieces exist in pairs.

5. Pattern making for a shirt — the sequence

1. **Take the measurement set** from the spec sheet (size M of Topic 2): chest 107, length 73, shoulder 46, sleeve 61.5, collar 40 (all cm, garment values).
2. **Draft the back** on the construction frame: length line, chest line at roughly one-quarter of chest below the shoulder line; shape the neckline, shoulder slope and armhole with curves; add the yoke line across the upper back and separate the yoke piece.

3. **Draft the front** from the same frame so front and back share chest and side-seam geometry; lower the front neckline, add the button placket extension (about 2 cm) and button stand, and place the pocket position by drill marks.
4. **Draft the sleeve:** cap height derived from the armhole (the sleeve cap perimeter must match the armhole with a small ease), biceps width from chest proportion, length to the spec value; shape the underarm seam.
5. **Draft the collar and cuff:** the band follows the neckline length (front + back neck curves walked and summed) at the collar size; the fall is drawn over the band with the collar's style shape; the cuff is a rectangle of wrist length plus overlap, width per style.
6. **Add allowances and markings** to every piece (§4.2–4.4), walk all seams (§4.5), write the information blocks, and cut out the production set.

6. Pattern making for a trouser — the sequence

1. **Measurement set** (example, size 32): waist 81, hip 99, inside leg 78, plus waist-to-hip depth about 24 cm (all garment values per spec).
2. **Draft the front:** frame of hip width; fix waist line, hip line and crotch (fork) depth; draw the fly curve from waist to crotch point; establish the **crease line** at the centre of the leg — all later width divisions are measured from it; shape knee and hem widths symmetrically about the crease.
3. **Draft the back** over the front's geometry: raise the back waist at the centre back (seat angle) so the garment sits without pulling, widen the seat, extend the back crotch point further for sitting ease, and add a waist dart.
4. **Draft the small pieces:** waistband (waist length plus extension), fly piece and shield, pocket bags and pocket facing pieces, belt loops.
5. **Allowances, markings, checking** as in §4: inseams and outseams of front and back walked against each other, crease lines aligned to grain.

The instructive difference between the two drafts is the crotch: the shirt is essentially a tube with holes, while the trouser must wrap two legs and a seat — its geometry, and its most common fit faults, all live in the curve from waist through the fork.

7. Pattern grading

DEFINITION

> Grading is the systematic increase or decrease of a master pattern to produce the other sizes of the range, by applying fixed increments (grade rules) at defined points of the pattern — so that every size keeps the style's proportions and fit character.

Grading does **not** mean enlarging the drawing by one percentage, as a photocopier would: the body does not grow uniformly — chest may step 5 cm while length steps only 2 cm and collar 2 cm. Each point of measurement has its own grade, taken from the spec sheet. For the shirt of Topic 2, the grade rules read directly off its POM table:

POINT	CHEST	LENGTH	SHOULDER	SLEEVE LENGTH	COLLAR
Grade per size (cm)	5	2	2	1.5	2

Measurement in size n = Base-size measurement + (number of steps × grade)

Worked example. The master pattern is size M (chest 107 cm). Size XXL is three steps up (M → L → XL → XXL), so chest = 107 + 3 × 5 = **122 cm**; length = 73 + 3 × 2 = **79 cm**; collar = 40 + 3 × 2 = **46 cm**. One step down, size S: chest = 107 − 5 = **102 cm** — exactly the spec table of Topic 2, which is how the table was built in the first place.

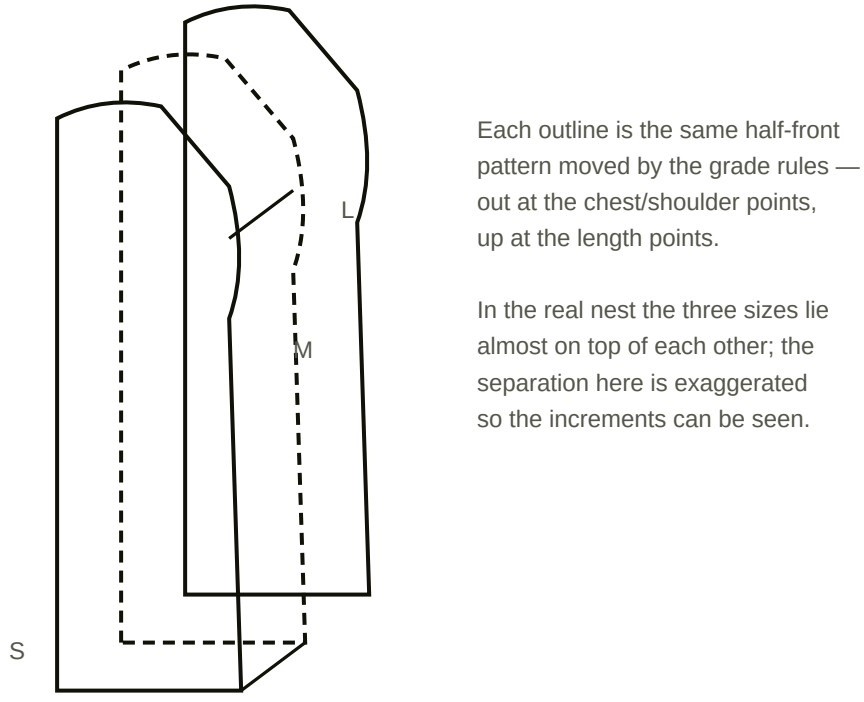


Figure 1 — A graded nest of the shirt front: sizes S, M and L produced from the master pattern by the grade rules (schematic — separations exaggerated).

Methods of grading. In **manual grading** the pattern is shifted on a ruled table by the x–y increments of each grade point and the new outline traced point by point (the stack/shift method). In **CAD grading**

the grade rules are stored with the master pattern's points and the computer generates every size instantly — the same rules, the same geometry, with none of the transcription errors. Either way, the graded set is checked by producing the spec table for all sizes and comparing it against the buyer's POM values.

8. Glossary

TERM	MEANING
Basic block / sloper	The plain, close-fitting master pattern drafted from body measurements, from which styles are developed.
Net (sewing) line	The line on the pattern along which the seam is sewn; allowances lie outside it.
Walking a seam	Measuring two sewing lines against each other to verify they join correctly.
Grade rule	The stored increment applied at a pattern point for each size step.
Grade point	A point of the pattern (shoulder tip, chest, hem…) whose position shifts during grading.
Nest	All graded sizes of one piece drawn superimposed, used to check the grading.

9. Key points and summary

- A pattern set is the physical form of the spec: block → working pattern → production pattern, each piece carrying allowances, grain line, notches, drill marks and an information block.
- Drafting follows a fixed construction order from the largest measurements; flat-pattern drafting is the industrial method, draping the couture one.
- Shirt drafting builds back, front, sleeve, collar and cuff on shared geometry; trouser drafting adds the crotch geometry and the crease line as its organising reference.
- Grading applies per-point increments from the spec — never a uniform enlargement — and is verified by rebuilding the all-size spec table.
- Checked, walked and complete: no pattern leaves the pattern room until its seams agree and its set is complete.

10. Practice

Answer in your own words; brief model answers follow.

1. Using the grade rules of §7, build the full spec row set for size XXL of the shirt (chest, length, shoulder, sleeve length, collar), taking M as base.

2. A trainee grades a pattern by photocopying it at 105%. Give two measurement errors this produces against the grade rules of §7.
3. List, in order, the pieces of a complete shirt production pattern set.
4. A front side seam measures 62.0 cm and the back side seam 63.4 cm. What does walking the seam tell you, and what are the two possible causes?
5. Why is the grain line marked on the pattern rather than left to the cutter's judgement at the table?
6. A trouser front is drafted with the crease line off-centre. Name two visible faults in the finished garment.

Answer key

1. XXL = M + 3 steps: chest 107 + 15 = 122 cm; length 73 + 6 = 79 cm; shoulder 46 + 6 = 52 cm; sleeve 61.5 + 4.5 = 66 cm; collar 40 + 6 = 46 cm.
2. Photocopying scales everything by 5%: chest would rise by 5% of 107 ≈ 5.4 cm (close by luck) but length would rise by 5% of 73 ≈ 3.7 cm against a rule of 2 cm, and collar by 2 cm by luck while shoulder rises 2.3 cm against 2 — the proportions distort because real grades are absolute and point-specific, not proportional.
3. Front ×2 (left/right fronts as one piece cut in pair or separate), back, yoke, sleeve ×2, collar band, collar fall, cuff ×2, pocket — each with allowances and markings, plus any interlining pieces specified.
4. The seams cannot join cleanly: 1.4 cm must be eased, stretched or will pucker. Causes: a drafting/measurement error in one panel, or designed ease in one seam that was not recorded — checking the spec decides which.
5. Because orientation relative to the warp decides hang, twist and pattern matching; the marker and lay are planned from the grain lines of all pieces together, and an error repeated across a lay ruins hundreds of garments, not one.
6. The crease will not hang at the centre of the leg (it drifts towards one side), and the two legs will twist unequally — the trouser looks rotated on the leg and the crease lines of front and back will not align.
